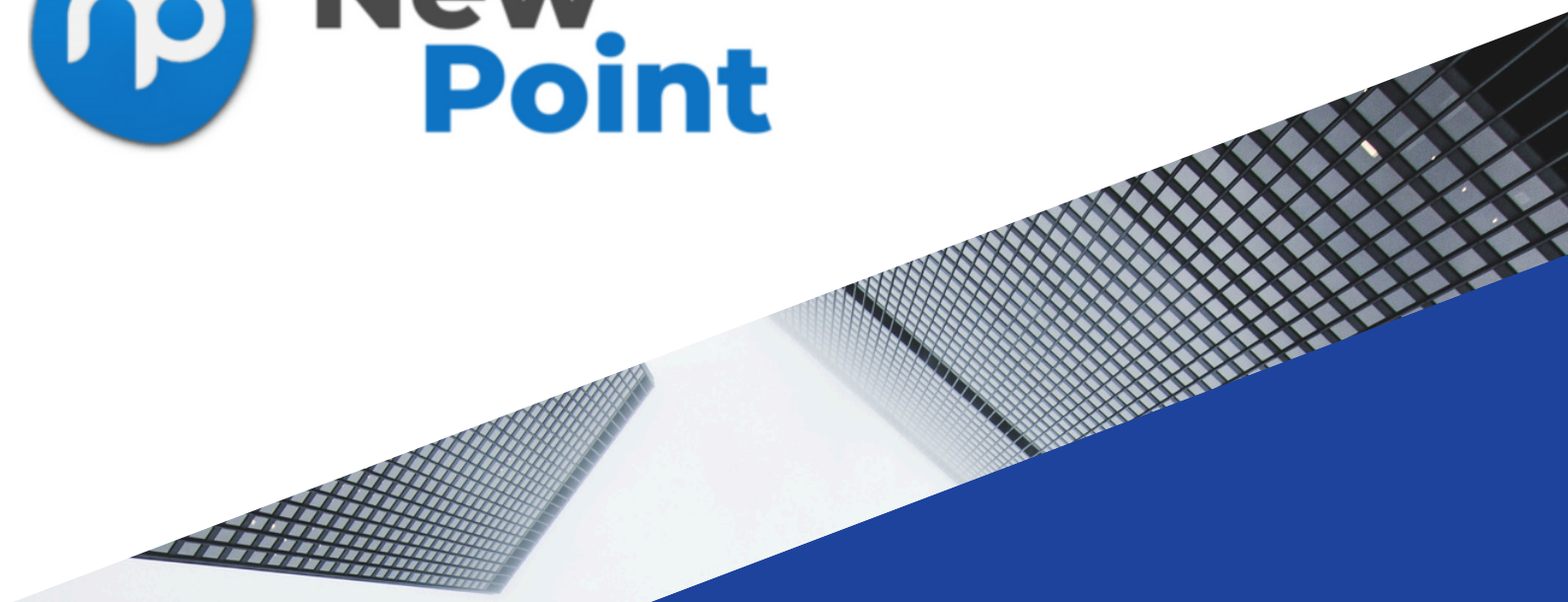
University: King Abdulaziz University.
College: Faculty of Economics and Administration
subject/department: MIS department
course: MIS 377 - System Analysis and Design

System Analysis and Design Report : Adding an Enhanced Feature to the NewPoint Accounting System for Al-Nahdah Auto Parts Company.

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New Point



Prepared by:

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Executive Summary

In this report, we will analyze the current status of Al Nahda Auto Parts' New Point accounting system, identify the most prominent shortcomings, and propose the best solutions to improve automation and accelerate system performance. The current system relies on manual processes for accounting, inventory management, and supplier communication, leading to slow price comparisons, delayed purchase orders, and consequently, wasted time, increased errors, and decreased productivity.

The proposed solution is to automate communication with suppliers when inventory levels are low. The system then receives and compares quotes, and proposes the optimal solution. This improvement ensures faster decision-making, data accuracy, and integration of financial and operational units.
Using the evaluation method, the automation feature received the highest score in terms of its effectiveness, its ability to reduce manual effort, enhance efficiency, and reduce errors resulting from manual performance, supporting the company's long-term digital transformation goals.

Introduction

In this project we are as a team planning to propose an enhancement for the existing accounting system which called NewPoint, many retail stores use it including the organisation that we will be writing about, they use it to manage their all financial processes and also they manage inventory and alot of processes using it. The goal of this enhancement is to address a critical issues that face alot of merchants, which include the manual and time consuming process because they need to keep track the inventory, identify products, communicating with suppliers and so on.

Al-Nahdah Company current situation is : the employer or the store owner must manualy contact suppliers and write every single thing they need and send it to the suppliers and compare them to get the best deal, and all of that is time and effort consuming.

The enhancement that we will be developing is : proposing a feature that will automate the whole workflow by sending automatic notifications to the suppliers that Al-Nahdah company dealing with whenever a shortage happen, then the suppliers can submit price offers through the system, and the merchant will be able to receive all offers in an interface and it will be easy for them to compare between them and select the best offer.

This report will give a detailed description for this enhancement and we will be applying the System Analysis and Design Concepts on it. The outline of the report is as following: Chapter 1 talkes in detail about the organisation and it's current system. Chapter 2 presents the needs assessment and problem analysis. Chapter 3 will discuss the proposed enhancement in detailed and mention the alternative solutions and evaluates them using appropriate techniques.

Chapter 1: Organizational Background

1.1 Introduction

This first Section of the report will provide an overview of Al-Nahdah Automotive Parts Establishment, it will include the company's background, vision, mission, goals, culture, and organizational environment. It will also describe the organisation structure and the current accounting system that the company using. and at the end there will be an ecosystem map and the SWAT analysis of Al-Nahdah Company.

1.2 Organizational Background

1.2.1 Organizational overview

Al-Nahda Automotive Parts Establishment it is a small company and was founded in 2021 in Jeddah, Saudi Arabia, they are specialized in selling the parts for alot of big and small companies and also for individuals, they are not just selling for customers inside Jeddah they also target customers from all over Saudi Arabia. The company gets its strength and competitive advantage through offering a high quality products and they are all original. The company also has built its reputation on honesty, trust, and product quality, ensuring its position in the market and to make the customer satisfied through receiving original and reliable product and also good service. Over the years, Al-Nahda was able to get good partnerships with multiple car part suppliers and local workshops to meet the increasing demand for authentic components.

The company in short believes that the customer is their main reason to survive and grow so they are trying their best to enhance their operational efficiency and serve them in the best way possible to get the customer satisfaction.

1.2.2 Vision:
to become one of Saudi Arabia's most reliable and cutting-edge providers of auto parts through establishing strong supplier and customer relationships and technology-driven operations.

1.2.3 Mission
To deliver original, premium auto parts to customers in a timely and effective manner while maintaining openness, dependability, and long term partnerships with suppliers and customers.

1.2.4 Goals and objectives:

Increase operational effectiveness by implementing automated systems.

Improve the company's contact with its suppliers.

Reduce the amount of time that inventory restocking takes.

Boost client satisfaction by providing quicker service.

Provide professional advice to help customers choose the right spare parts.

1.2.5 Culture

The company’s culture is to build integrity, honesty, quality, and make a continuous improvements to face the dynamic environment.holding honesty is one of their important priorities that their employees are encouraged and trained really well for that, they are trained to serve the customer by not just selling but also giving advices and serving them fast. The organization values trust and transparency both internally and externally.

1.2.6 EEF

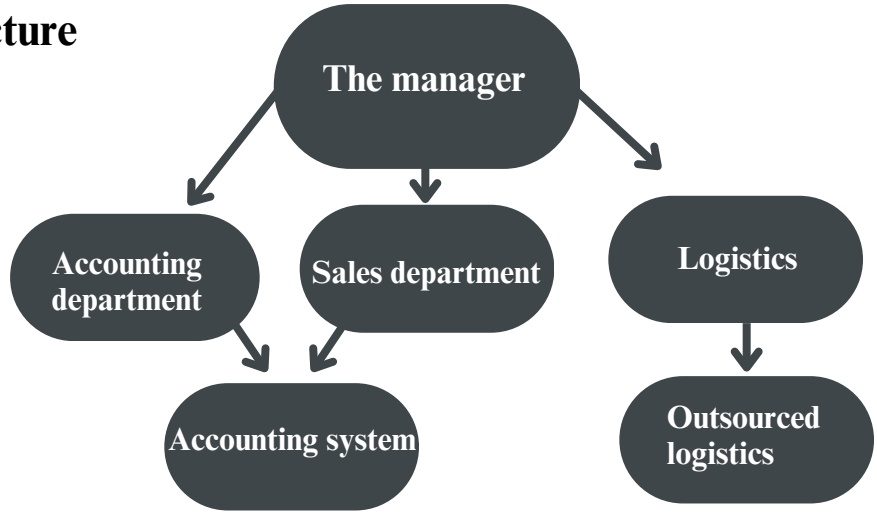
Community:
The location of the store of Al-Nahdah company is very strategic and busy commercial area with alot of stores working on the same category like it, that's considered good and bad at the same time, it's disadvantage because the competition increases and that makes it very difficult to differentiate in the market, but the good thing is the easy access for customers and suppliers. The community or customers include individuals with different ages and also different education levels, but many customers have basic automotive knowledge. Income levels vary, but most of them are from the middle class that prefer affordable products with good quality.

Economic:
The market for auto parts in Saudi Arabia is strong because the number of vehicles is considered high. Even though there's a strong competition, but the demand is stable in general during the year.

Political:
There is a good political stability and that enhances the business environment and increases opportunities for expansion.

Legal:
The company has alot of responsibilities, it must follow Saudi laws and regulations. The most important laws are: comply with religion, professional ethics, commercial documents, registration in the commercial register, and registration with the Chamber of Commerce and Industry.

1.3 Organizational structure



1.4 Organizational Systems:

Al-Nahda Auto Parts Store mange their operations using a single accounting system, this system is very important for daily operations. The main use of this system is for managing purchase orders from suppliers, record orders and handle accounting tasks. The system supports efficient communication with suppliers in addition to its primary role for internal management. And that helps the company's store maintain smooth operations.

1.3.1 current system

The store now manages inventory and purchases using a typical accounting system. Even though they have a strong accounting system but They still inside the system manually determines and know that there's a shortage and gets in touch with each supplier to ask for prices as supply runs low. Due to the need to manually record shortfalls, submit inquiries, and compare supplier responses, this process take alot of time and bring mistakes.

1.3.3 SWOT

Strengths

- Good supplier network.
- Experienced staff.
- Convenient location.
- Delivery available

Weaknesses

- Small team.
- Dependence on one system.
- Limited marketing.

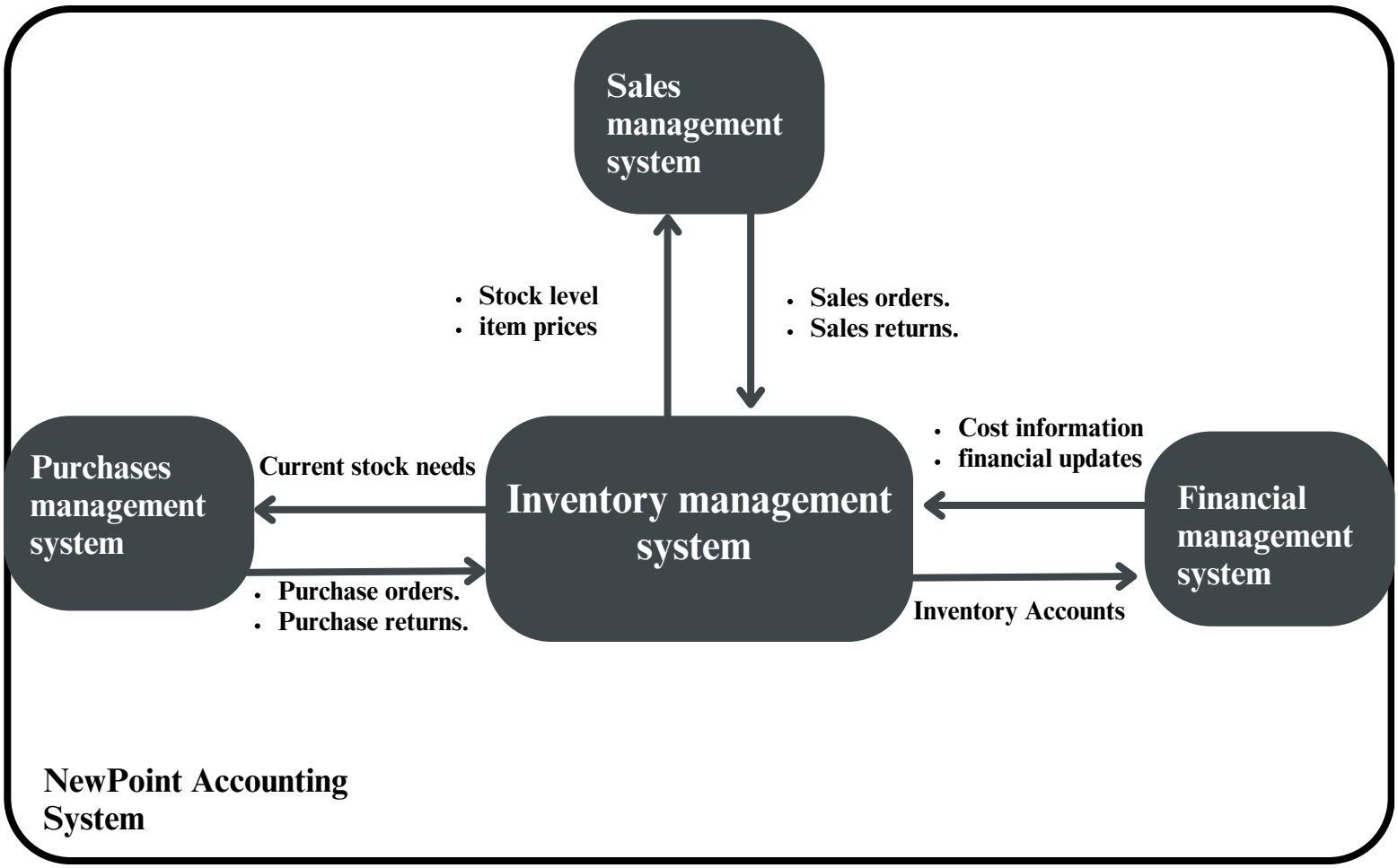
Opportunities

- Growing car market.
- Online sales potential.
- More supplier options.

Threats

- Strong competition.
- Changes in customer preferences or car models.
- Increasing use of electric cars over time.

1.3.2 Ecosystem Map



1.5 Conclusion

In conclusion, this chapter gave a summary of alot of things regarding Al-Nahdah Company, and presented its accounting system. Now it will be easier to identify the organization's primary issues and get ready for system enhancement in the upcoming chapters when one is aware of the organization's surroundings and structure.

Chapter 2: Needs Assessment

2.1 Introduction

In this chapter, we will discuss the needs of the current accounting system used by Al Nahda Foundation. This assessment includes defining the problem, conducting a gap analysis, and describing the project scope. It then goes on to analyze the challenges, identify objectives, and identify key requirements for improvement. One of the most prominent systems currently deployed in the retail sector is the New Point accounting system. However, operations management faces many challenges in communicating with suppliers and processing purchases efficiently. The process remains manual and time-consuming, potentially leading to delays. This delay leads to increased errors and, consequently, decreased productivity.At the beginning of the chapter, the main problems facing the current system and their causes are explained. After that, the objectives and constraints related to development and improvement are clarified. Then, the gap analysis is presented in a table that shows the most important differences between the current situation and the future situation. Finally, this chapter will define the scope of the project and the stakeholders associated with it.

2.2 Current system assessment

There are many challenges facing the NewPoint accounting system used by Al Nahda Foundation, which affect the daily operations. The system is one of the important things that contribute to managing financial transactions and controlling inventory. However, when automation is absent in communicating with suppliers, the process becomes cumbersome and time-consuming. This limitation may reduce the efficiency of the system and cause delays in replenishing essential items.

Chapter 2: Needs Assessment

5 Why Analysis

1- Why does the purchasing process take so long?

- Because employees have to manually communicate with suppliers and wait for their responses to obtain quotations.

2- Why do employees communicate with suppliers manually?

- Because the system cannot send automatic notifications when inventory levels run low.

3- Why doesn't the system send automatic notifications?

- Because there is no automation feature for communicating with suppliers and monitoring inventory.

4- Why doesn't the current system have automation?

- Because it was designed specifically for accounting purposes without operational integration.

5- Why hasn't the system been updated to include automation?

- Because most of the company's coding was focused on managing day-to-day operations rather than developing and improving the system.

Chapter 2: Needs Assessment

2.2.1 Problem Definition

Problem Statement:

The current problem with the system is that when the stock drops to the lowest level, the suppliers are not automatically notified and the employees are required to manually carry out the purchasing process step by step, starting from identifying the shortage to arrival and communicating with the suppliers, discussing and comparing prices. This process causes a delay in the stock to be refilled and places a heavy burden on the employees and puts them under constant pressure from running out of essential products.

Issues:

1- The system requires manual inventory tracking, which leads to slow response to stock shortages.

2- Employees communicate with suppliers individually, which consumes a lot of time and effort.

3- Price quotations reach different channels such as (mail, text messages, or calls), which makes employees distracted and difficult to track.

4- There is no unified comparison interface, which makes it difficult for employees to compare prices manually.

5- The lack of automation has led to delayed purchases, weak communication, and division between departments.

Objectives: **Weight (1 - 10, with 10 being the most important)**

1- Automate the process of communicating with suppliers when stocks run low.(10)

2- Enable the system to track orders automatically and detect shortages in real time. (9)

3- Develop a dashboard within the system that compares suppliers' offers. 8

4- Reduce the purchasing cycle time and also reduce human intervention. 8

5- Improve data accuracy and enhance decision-making efficiency.7

Constraints:

1- The project cost should be 20,000 Saudi Riyals.

2- The improvements should be made and fully operational within six months.

3- The new improvement should be integrated into the current structure of the New Point system without affecting accounting tasks.

4- The new automation feature should not cause any problems or conflict with the core system modules.

2.2.2 GAP Analysis

Current State	Desired State	Gap
Inventory checks are performed manually by employees.	A system that automatically detects declines and sends notifications to suppliers.	There is no automated inventory control
Suppliers are contacted individually.	The system automatically sends notifications to multiple suppliers simultaneously	There is no automation in communication.
Quotations are collected manually via mail and phone calls.	Bids are sent and managed digitally within the system.	There is no unified platform for offers.
Quotations are compared manually.	An automatic comparison dashboard is created in terms of (price, delivery time, quality).	There is no analysis and comparison tool.
Delays in creating purchase orders.	The system automatically generates purchase orders after the bid is approved.	There is a lack of integration between units.

Table 2.1: Gap Analysis

2.3 Scope Description

The proposed improvement at Al Nahda Foundation focused on developing the Newpoint system by adding new automation features that will facilitate communication with suppliers and inventory management. This update will also allow the system to automatically detect issues related to low stock levels and send immediate notifications to suppliers, ensuring faster replenishment and increasing operational efficiency.
In addition, the improved system will include a unified interface that allows suppliers to submit quotes directly, allowing store owners to easily compare quotes and create purchase orders instantly. This improved system will help reduce manual work and errors, and contribute to improving and developing the overall operational performance within the organization.

2.4 Stakeholders Identification

ID	Name	Title	Role	Category	Contant	Communication Requirements
1	Bassem	Project owner	Supervise project	Internal	Email or Number	Supervises the company and reviews progress weekly.
2	Bassem/ employees	Stock Officer	Monitor stock	Internal	Email or Number	Sends daily stock reports and communicates any shortage issues
3	Nizar Ali	System Developer	Maintain system	Internal	Email or Number	Communicates when issues appear and provides regular system updates
4	Redwan	accountant	Review finance	Internal	Email or Number	Reviews financial data and accounts and submits a weekly report.
5		Suppliers	Send quotations	External	Email or Number	They receive notifications and submit quotes.
6		Customer	End Users	External	Email or Number	Place orders and provide feedback when needed

Table2.2: Stakeholders Identification

2.5 Conclusion

In this chapter, we analyzed the problem of the current accounting system of Al Nahda Company and used the 5 Why technique method. We started by detailing the problem, setting its objectives and issues, and setting the priority of the measure for each objective. Then, we identified the gaps and described the scope. Finally, we showed the areas of need for automation and identified the scope of improvement and the main stakeholders to ensure communication and efficient project management.

3.2.1.1 Solution Capability Matrix

Capability	Current State	Desired State	Gap
Supplier communication	Manual contact by employees	Automatic notifications and quotations	Lack of automation
Quotation comparison	Manual review of supplier offers	Automatic dashboard for ranking and evaluation	No digital comparison tool
Inventory tracking	Checked manually	Real-time tracking and alerts for low stock	No automated monitoring
Purchase order generation	Manual entry in system	Automatic creation after approval	Lack of integration
Data management	Backlog of emails and spreadsheets	Centralized and synchronized database	Limited data organization

Table3.1: Capability Matrix

Chapter 3: Alternative Analysis

3.1 Introduction

This chapter discusses alternatives for improving the New Point accounting system. Improving communication with suppliers is the main objective of this chapter. Three workable solutions are available, each designed to improve the company's operations and address its problems and constraints. We used the solution possibilities matrix and the weighted scoring technique to evaluate each option and identify the alternative. The chapter also explores the feasibility study for future assurance from a technical and operational perspective.

3.2 Proposed systems

3.2.1 Systems description

Solution 1: Automated Supplier Communication Feature

The first solution seeks to integrate an automated supplier communication module into the New Point system. This feature automatically detects low inventory levels and sends notifications to registered suppliers who request quotations. Suppliers can respond via the system portal, and employees can compare them after receiving all quotations.
The solution provides a dashboard to allow employees to compare quotations and categorize suppliers based on prices and delivery times, automatically arranging them. By automating these two processes, the company saves time and reduces human error, taking the company to the next level.

Overall, this solution enhances operational efficiency, seeks to improve supplier relationships, and ensures better, timely decisions are made based on data.

Solution 2: Separate database for inventory and purchasing management

An independent database for purchasing and inventory management. This solution proposes creating an independent inventory and purchasing management system separate from the NewPoint system. This system manages supplier data, bids, and purchase orders, and allows for continuous tracking of inventory levels. Employees input supplier data, community categories, and reorder points, and the system then automatically generates a request for quotation (FFQ) and sends it to approved suppliers.
This solution tracks inventory in real time and provides accurate data on parts availability. The system can also be linked to the NewPoint system to synchronize financial data. The disadvantages of this solution are that it requires additional setup and higher maintenance costs, but it provides greater flexibility and the possibility of future expansion as the business grows.

Solution 3: Maintain the current system.

The third solution proposes maintaining the system as is, continuing to use the same processes when communicating with suppliers, comparing bids, and managing purchase orders. The most significant advantages of this solution are its low cost and the lack of need for technical updates. However, it will not address any of the fundamental issues currently facing the organization, such as slow processes, decreased efficiency, and increased errors.
In the long run, this solution will lead to slower workflow, loss of competitiveness, and increased operating costs. Therefore, it is not recommended to pursue this solution because it does not offer any real improvement to the current system.

3.2.2 Weighted Score technique

This table uses a weighted scoring tachnique to evaluate the three proposed solutions for developing Al-Nahdah Foundation's system. Each alternative is evaluated based on four main criteria: **cost, ability to solve the problem, development risks, and time to implement**. Each criterion is then weighted according to its importance. The weight is multiplied by the assigned score to obtain a weighted score. The total score is then used to determine the most feasible and efficient alternative.

Project	Criteria	Weight	Score	Weighted Score
Automated Supplier Communication Feature	**Cost**	**2**	**3**	**6**
	Solve problem	**3**	**4**	**12**
	Development risk	**2**	**3**	**6**
	Time to implement	**1**	**3**	**3**
			Total Score	**27**
Separate database for inventory and purchasing management	**Cost**	**2**	**2**	**4**
	Solve problem	**3**	**3**	**9**
	Development risk	**2**	**2**	**4**
	Time to implement	**1**	**2**	**2**
			Total Score	**19**
Maintain the current system	**Cost**	**2**	**3**	**6**
	Solve problem	**3**	**1**	**3**
	Development risk	**2**	**3**	**6**
	Time to implement	**1**	**1**	**1**
			Total Score	**16**

3.2.3 Recommended solution

Based on the results, the first solution—the automated supplier communication feature—was the most effective option.

This solution offers the highest level of integration with the existing NewPoint system, while improving supplier interaction and inventory control. It enables real-time tracking, automatically manages quotes, and speeds up purchase order creation, enhancing the efficiency and accuracy of business processes. By automating communication and consolidating supplier responses into a single dashboard, this significantly reduces time and effort for the company and enhances decision-making with reliable and up-to-date information.

Chapter 3: Alternative Analysis

3.3 Feasibility Study

3.3.1 Operational Feasibility

The operational feasibility of adding an automated supplier communication module to the NewPoint accounting system is high, as it will facilitate supplier interaction and purchasing processes. Manual monitoring and communication, which currently cause delays and errors, will be eliminated. The new system automates monitoring by detecting low stock levels, sending RFQs, and organizing supplier responses into a single portal. This system saves time, reduces human error, and ensures the accuracy of information. Employees can easily compare supplier performance in terms of price and delivery through a central dashboard, speeding up decision-making and strengthening their relationships with suppliers. Automation also allows employees to focus on strategic tasks such as cost analysis and forecasting, improving overall productivity. Additionally, the system provides a clear audit trail, enhancing accountability and transparency. Easily integrated into existing workflows, the solution enables more efficient operations, timely procurement, and fact-based decision-making, enhancing business performance and supplier collaboration in the long run.

3.3.2 Technical Feasibility

The technical feasibility of adding an automated supplier communications module to the NewPoint accounting system is highly feasible. Because the system is modular and scalable, new functionality can be added without impacting daily operations or data integrity. The module will connect via secure APIs to receive inventory, supplier, and purchasing data, leveraging existing infrastructure and widely supported technologies, such as automated notifications and dashboards. Integration will be easy with minimal configuration and a reasonable cost, given that the database and servers are already in place. Strong security measures, including encryption and role-based access, will ensure the protection of supplier data. Implementation will be a systematic process involving testing and training to ensure a smooth transition. Maintenance requirements will be minimal, and future enhancements, such as predictive analytics, will be easily added. Overall, this solution is flexible, cost-effective, and technically amenable, enhancing system efficiency and contributing to the company's long-term technological growth.

Chapter 3: Alternative Analysis

3.5 Conclusion

In conclusion, integrating the automated supplier communication module into the NewPoint accounting system is the optimal and most effective solution. Compared to other alternatives, this system offers superior operational and technical advantages, improving supplier communication, reducing manual workload, and ensuring data accuracy. It also enhances decision-making through automation, real-time tracking, and centralized dashboards. Technically, the system is scalable, secure, and easy to integrate, leveraging existing infrastructure with minimal cost and downtime. Overall, this solution supports sustainable growth, operational efficiency, and a data-driven approach to supplier management.

Conclusion

In conclusion, the proposed enhancement to Al Nahda Auto Spare Parts' New Point accounting system effectively addresses the major operational challenges caused by manual supplier communication and inventory management. By incorporating an automated supplier communication module, the system will streamline purchasing processes, reduce errors, and improve decision-making efficiency. This automated system supports faster responses to low inventory levels and facilitates seamless quote comparisons, saving time and costs. This practical solution is technically and operationally aligned with Al Nahda's mission of reliability and innovation, enhancing its competitive advantage. Finally, this enhancement will enhance productivity, transparency, and sustainable digital transformation across the company.

Resources

نيو بوينت من افضل واسهل البرامج المحاسبية ونقاط البيع. (n.d.). /https://newpointerp.com

https://www.gep.com/blog/strategy/procurement-automation-meaning-benefits-examples

https://www.mdpi.com/2071-1050/13/14/7520

Direct communication with Al Nahda Company was conducted to gather information.